Audio

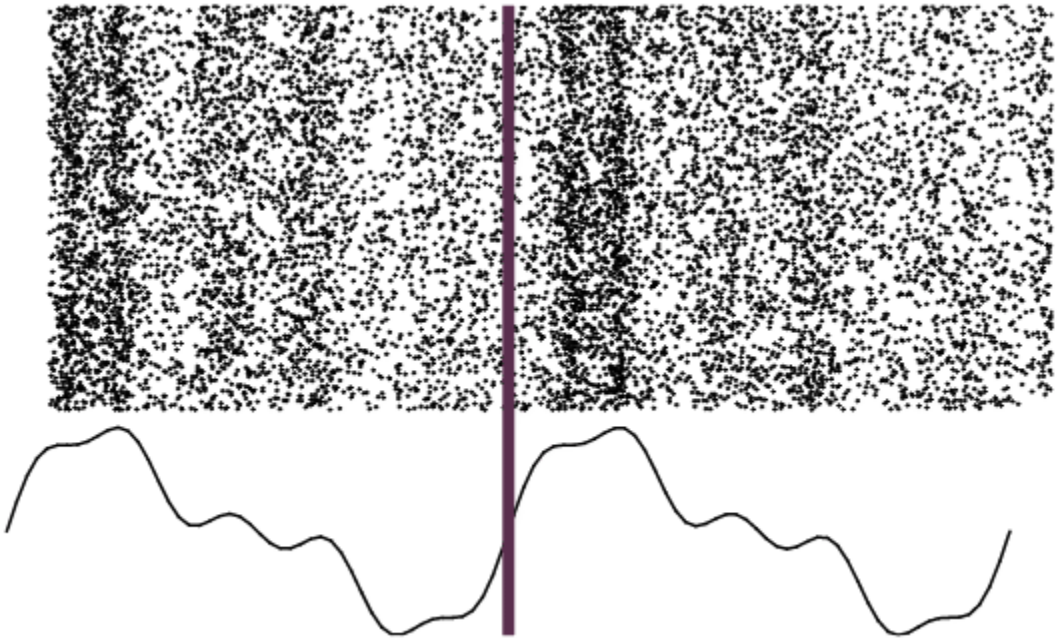
Dr. Josef Fruehwald

The Upshot

Recording sound and doing acoustic analysis is all about managing trade offs.

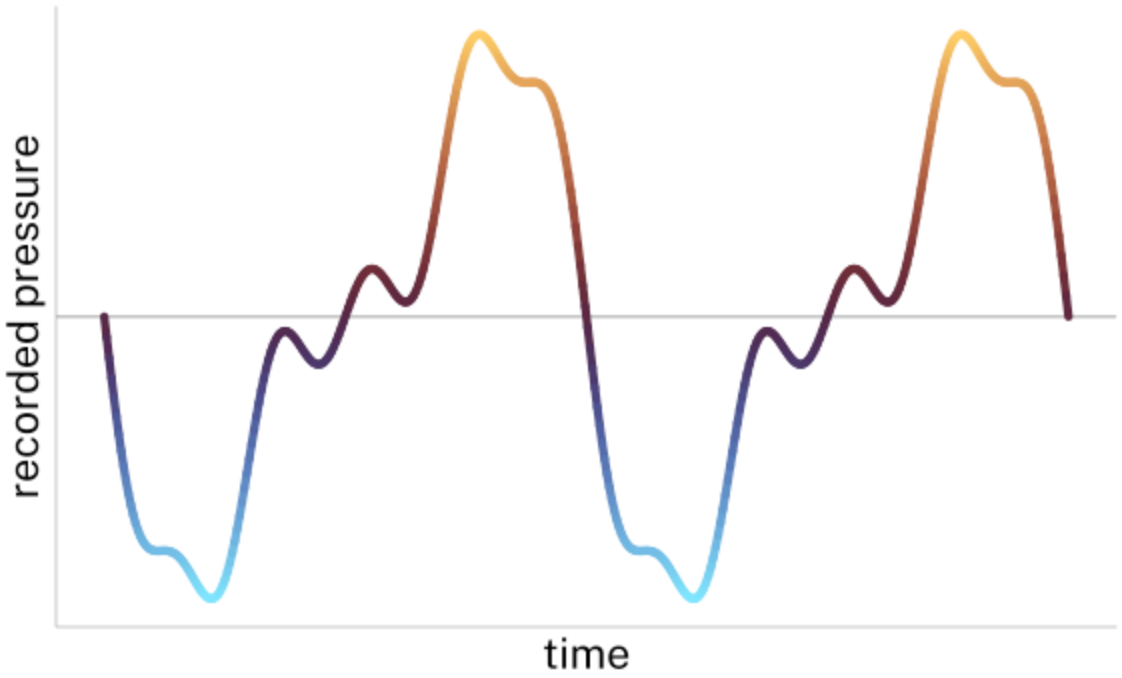
Sound

From space:

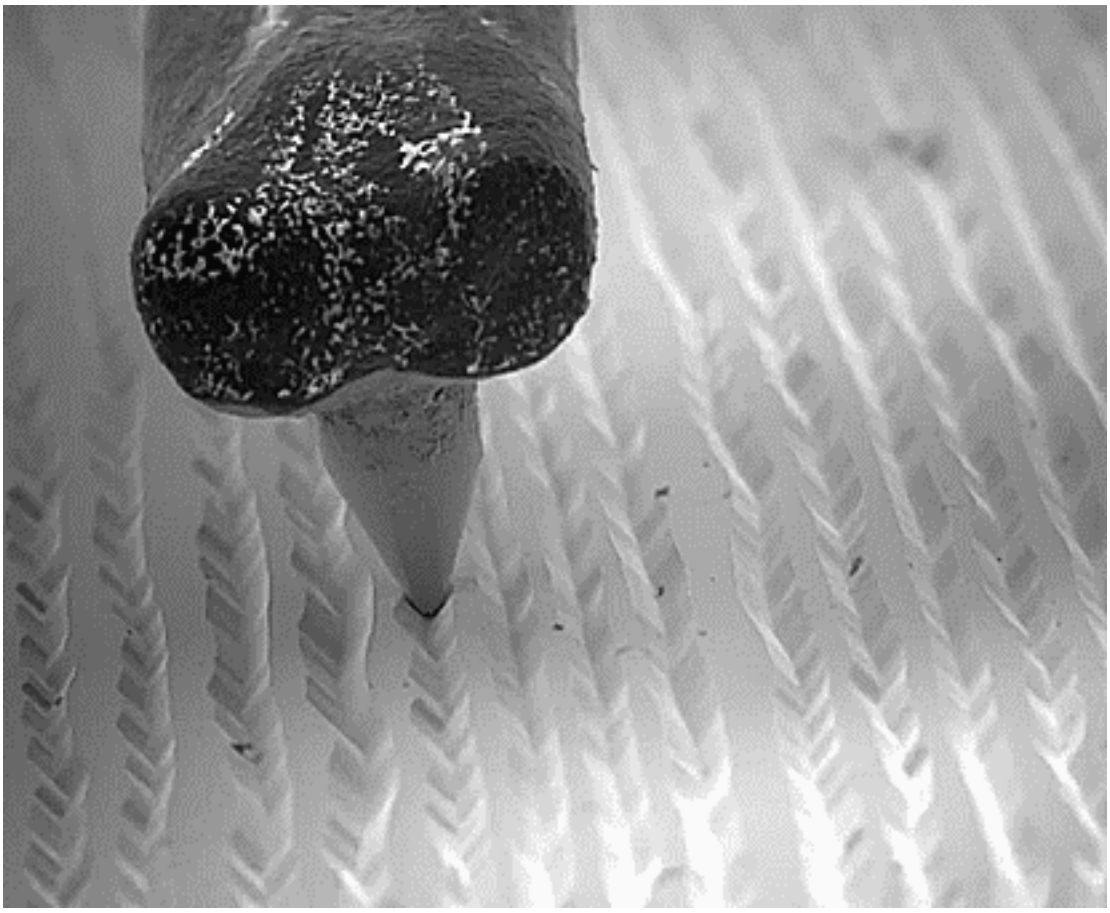


Sound

To time:


recorded pressure
time

Analog Audio



Analog Audio







Analog Audio

YouTube

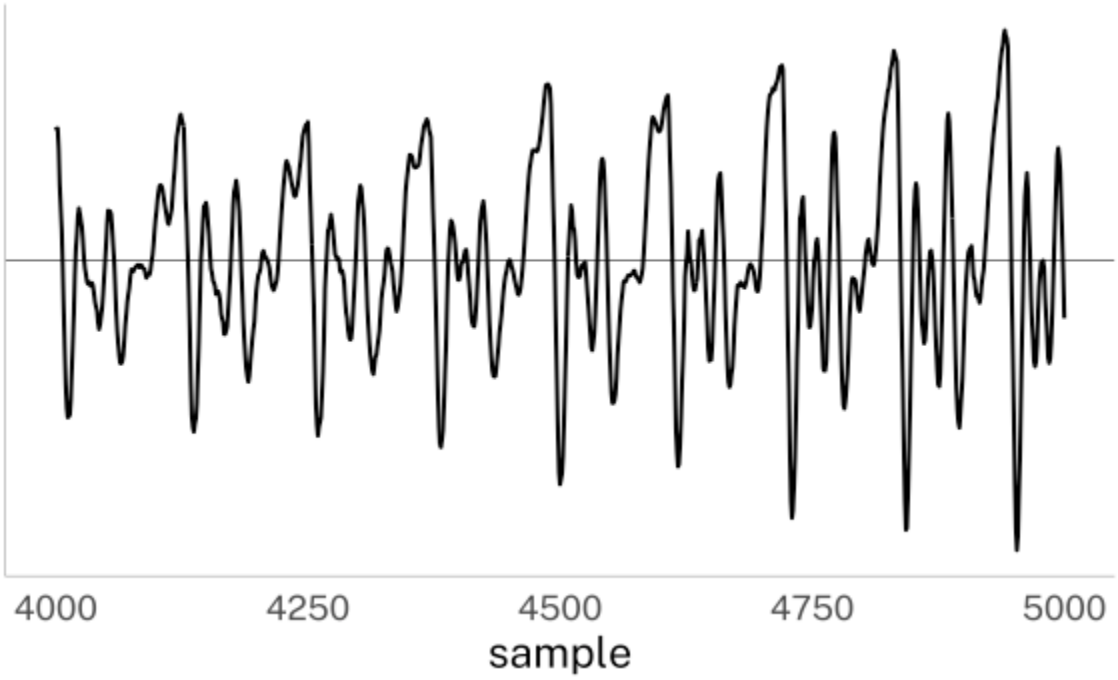


Digital Audio

Instead of etching continuous grooves, we can note down a sequence of high and low pressure values.

```
library(tuneR)
wav <- readWave(here::here("data/audio/harmonics.wav"))
wav@left[4000:4010]
```

```
 [1]  4640  4567  4592  3869  2631  1848  1090  -239 -1731 -3085 -4311
```



Digital Audio

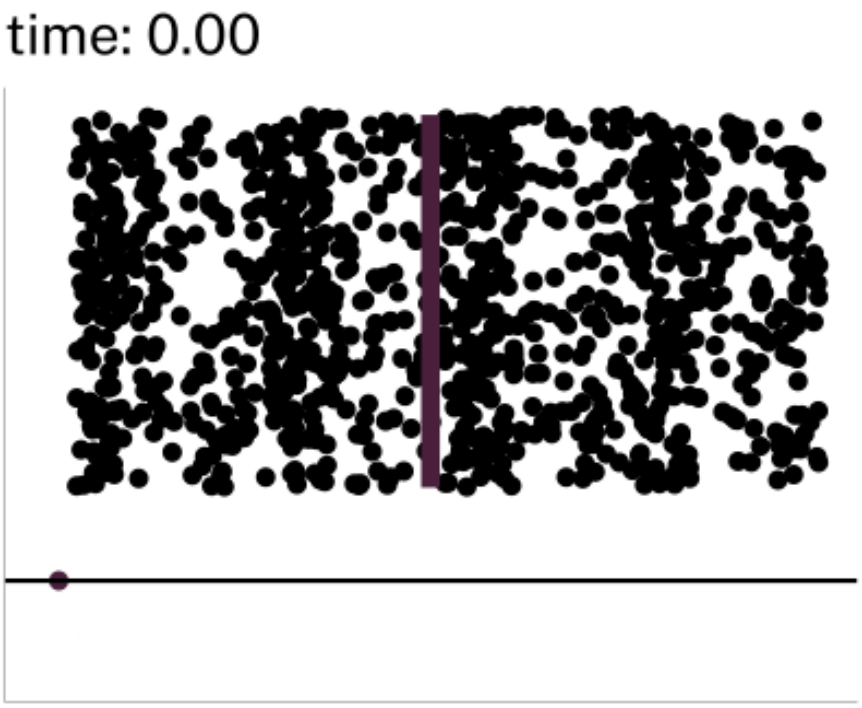
Let's say we decide to note down that the air pressure at the first moment in the recording was −194.

When do we decide that the *second* moment is, to write down the *second* air pressure?

Time doesn't come in discrete moments.

Digital Audio

Sampling rate



Sampling Rate

- Described in Hz.
- 16,000 Hz sampling rate = 16,000 samples per second

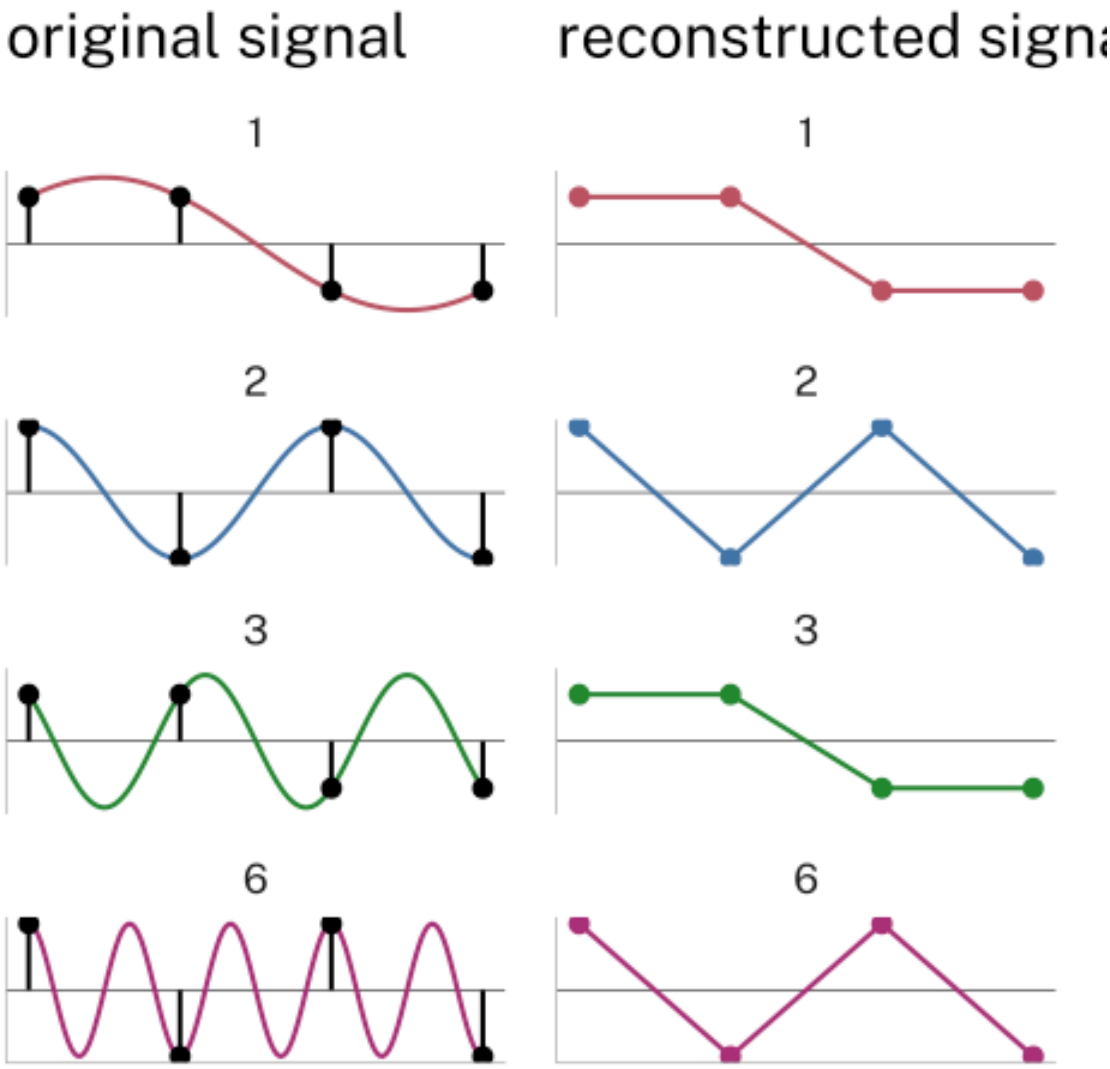
Sampling Rate

Common Sampling rates

Rate	Context
8,000 Hz	The sampling rate on telephones.
16,000 Hz	Commonly used in speech processing.
44,100 Hz	Standard sampling rate for MP3s
48,000 Hz	Standard sampling rate for professional digital recording

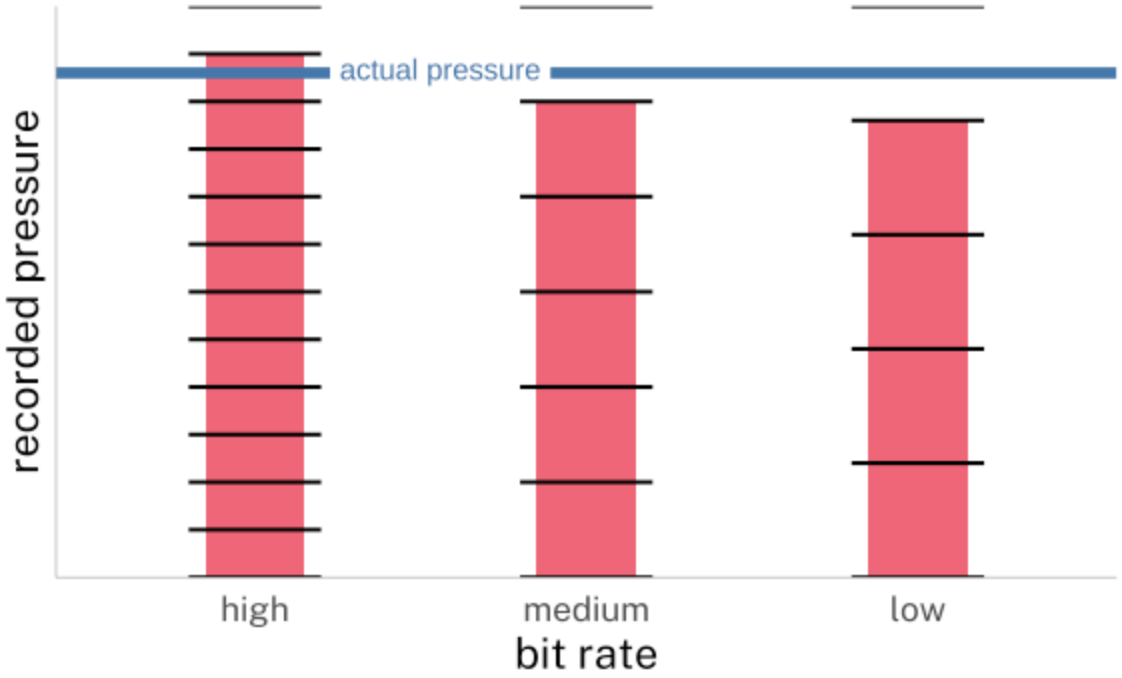
Sampling Rate

You can record sound frequencies up to $\frac{1}{2}$ the sampling rate (the Nyquist Frequency).



Bit Rate

Computers can only record numbers with finite precision.



Sample and Bit Rate

Why not always max out sampling rate and bit rate?

Precision & File Size

Duration
0.89s

Sampling Rate
44100 Hz

File Size
77.1K

If it was 1 hour
305M

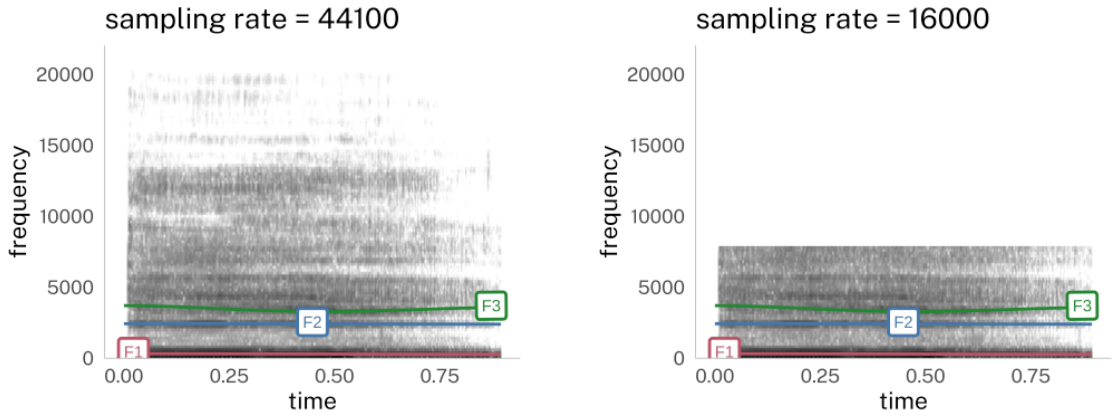
Duration
0.89s

Sampling Rate
16000 Hz

File Size
28K

If it was 1 hour
111M

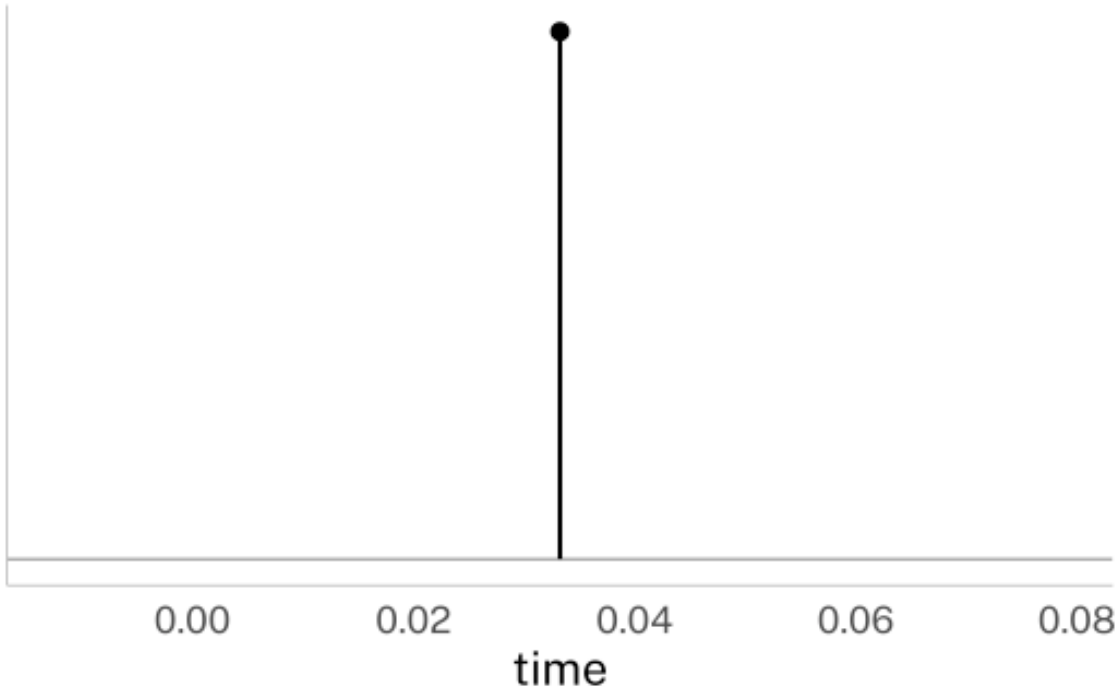
What's really necessary?



Acoustic analysis

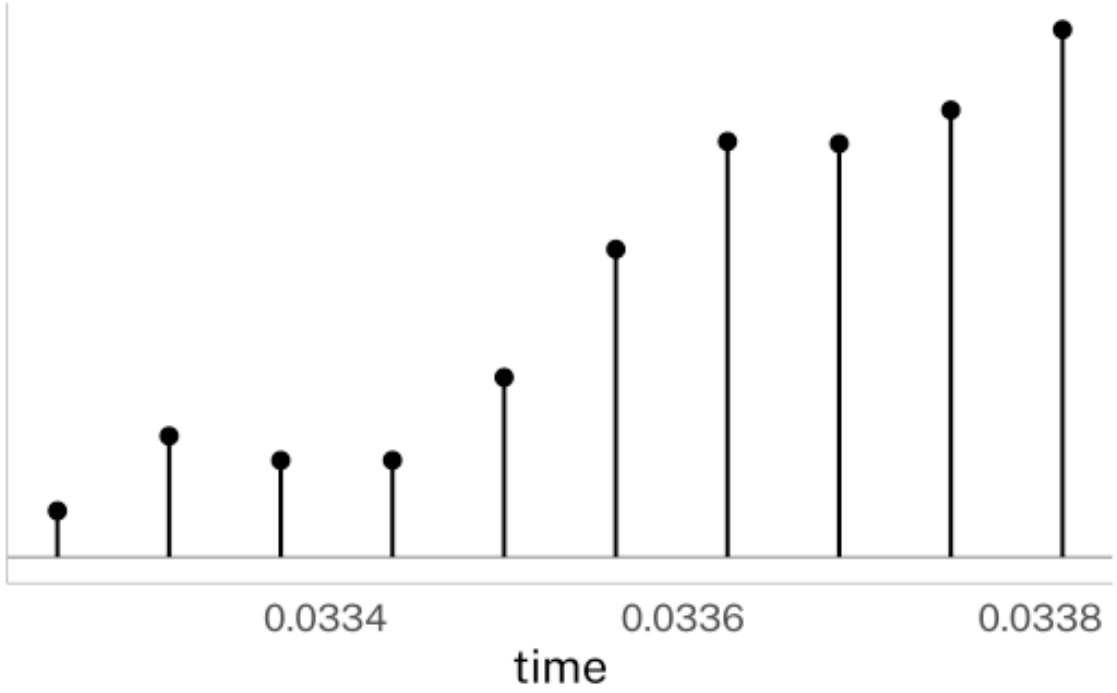
Let's say you've decided on a sampling rate, and now you want to estimate the fundamental frequency.

At a single point in time, we have a single sample. At what frequency is the signal alternating?



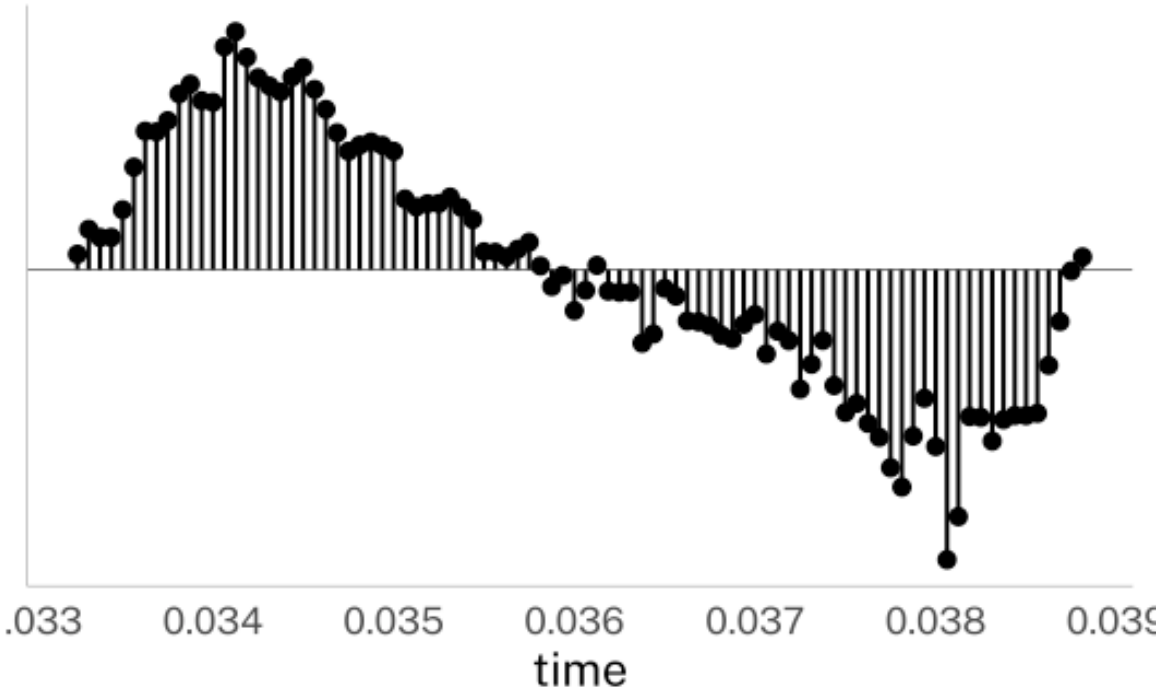
Acoustic Analysis

Ok, here's 10 points.



Acoustic Analysis

It looks like we need 90 samples or $\frac{90}{16000}$ = 0.005625 seconds to get a single period.

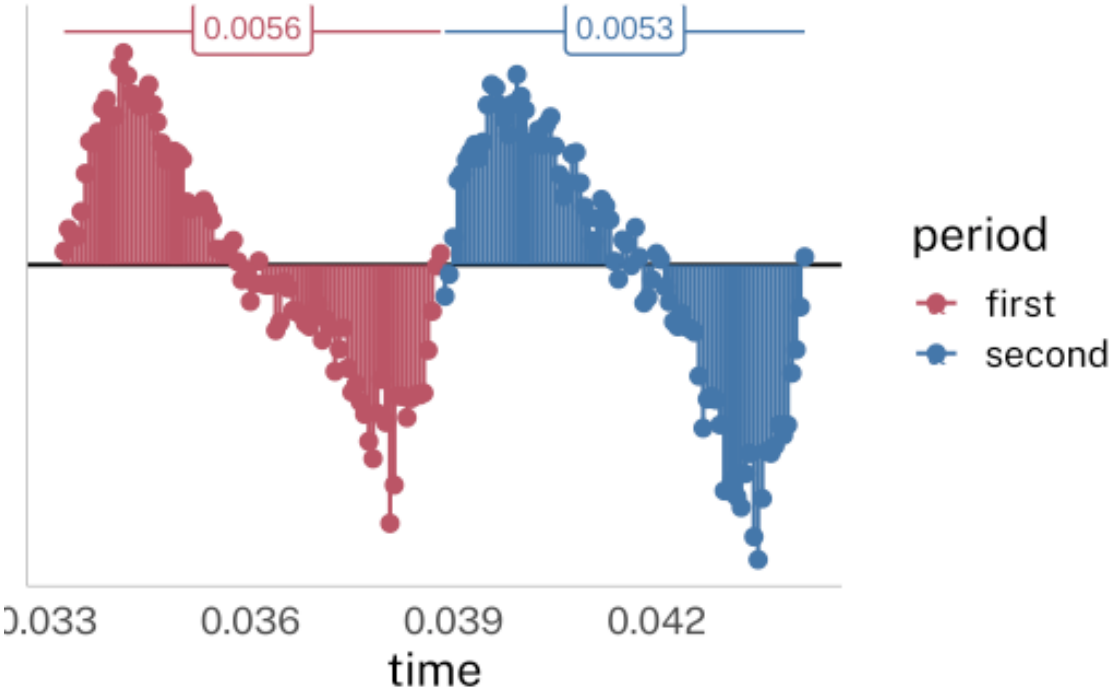


Acoustic Analysis

Analyses of frequency can only be done over spans of time. We always need to choose an analysis window size.

Acoustic Analysis

The second period isn't exactly the same.



Acoustic Analysis

period	samples	duration (s)	frequency (Hz)
first	90	0.00562	178
second	86	0.00537	186

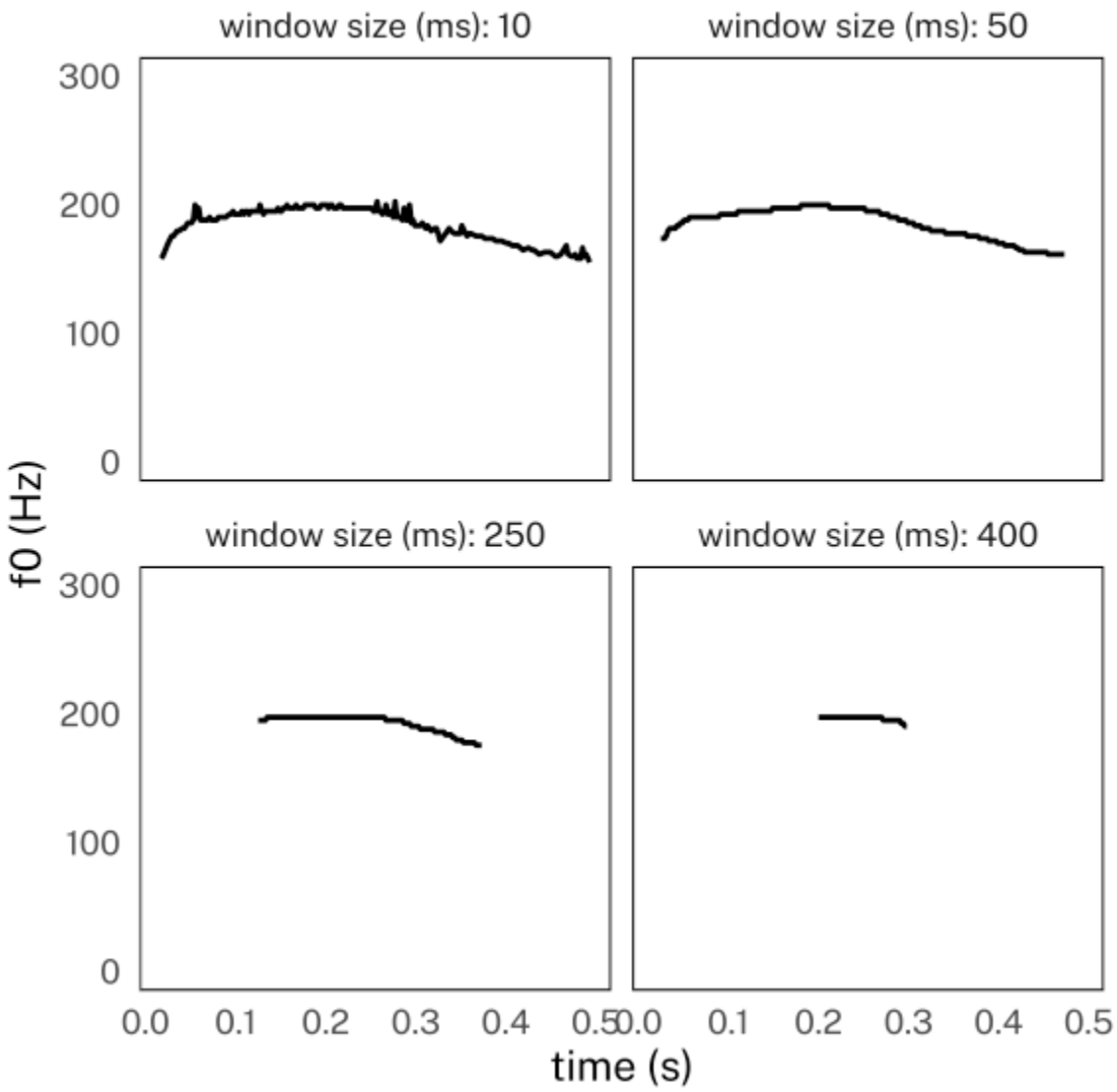
Acoustic Analysis

What if we expanded the analysis window to the entire vowel for greater accuracy?

time (s)	frequency (Hz)
0.447	176

But... the way F0 changed over the course of the vowel has been lost.

Acoustic Analysis

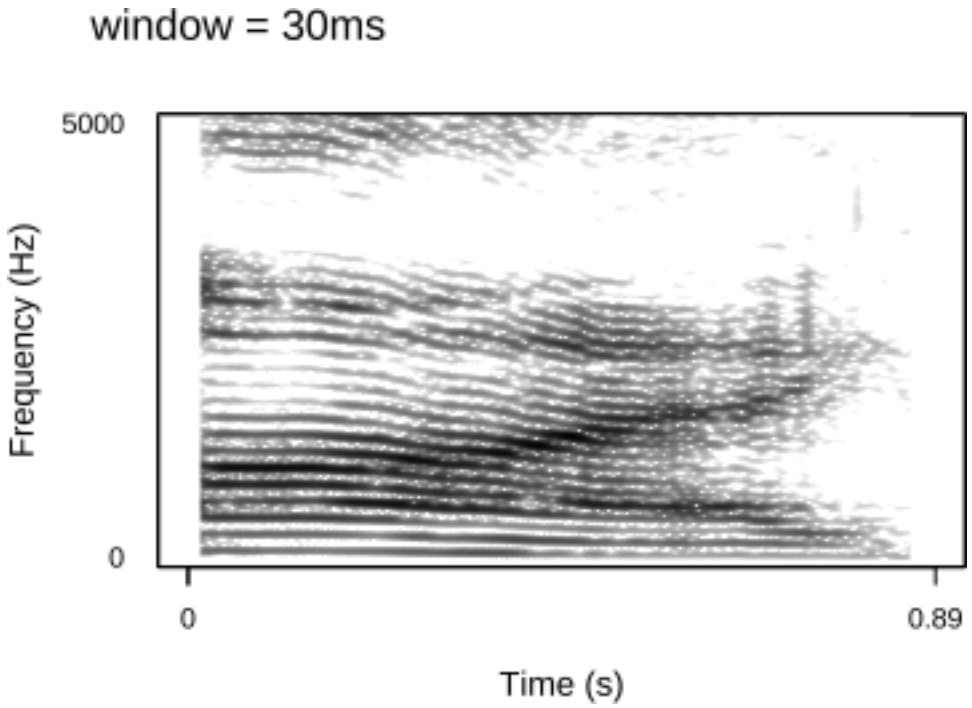


Acoustic Analysis

These same trade off between time and frequency needs to be made with spectrograms as well.

Acoustic Analysis

Narrow band spectrogram = wide analysis window



Acoustic Analysis

Wide band spectrogram = narrow window

